

2N7002

2N7002 — N-Channel Enhancement Mode MOSFET
Manufacturer: Diodes Incorporated Package: SOT-23

General Description

The 2N7002 is an N-channel enhancement mode MOSFET with a 60 V drain-source rating in a SOT-23 package. Suitable for low-power switching, level shifting and relay driving.

Features

- Drain-source voltage $V_{DS} = 60\text{ V}$
- Continuous drain current $I_D = 300\text{ mA}$
- Low on-resistance $R_{DS(on)} = 5\text{ ohm}$ at $V_{GS} = 10\text{ V}$
- Gate threshold voltage 1.0 to 2.5 V

Pin Configuration

Pin assignment for the SOT-23 package (top view).

Pin	Name	Function
1	Gate	Control input
2	Source	Source, usually grounded
3	Drain	Switched output

Absolute Maximum Ratings

Stresses beyond these values may cause permanent damage to the device.

Symbol	Parameter	Value	Unit
V_{DS}	Drain-source voltage	60	V
V_{GS}	Gate-source voltage	± 20	V
I_D	Continuous drain current	300	mA
I_{DM}	Pulsed drain current	800	mA
P_{tot}	Total power dissipation	300	mW
T_j	Junction temperature	150	degC

Electrical Characteristics

Typical values at TA = 25 degC unless otherwise noted.

Symbol	Parameter	Test condition	Min	Typ	Max	Unit
VGS(th)	Gate threshold voltage	VDS = VGS, ID = 250 uA	1.0	1.8	2.5	V
RDS(on)	Static drain-source on-resistance	VGS = 10 V, ID = 500 mA	-	3.0	5.0	ohm
IDSS	Zero gate voltage drain current	VDS = 60 V	-	-	1.0	uA
Qg	Total gate charge	VDS = 10 V, VGS = 10 V	-	0.6	1.5	nC

Thermal Characteristics

Symbol	Parameter	Value	Unit
RthJA	Thermal resistance junction-ambient	417	K/W

Package Dimensions

Outline drawing for the SOT-23 package. All dimensions in millimetres.

Dimension	Min	Nom	Max	Unit
A (body length)	2.80	2.90	3.04	mm
B (body width)	1.20	1.30	1.45	mm
C (height)	0.89	1.00	1.12	mm

Ordering Information

Order code	Package	Marking	Packing
2N7002	SOT-23	702	Reel, 3000 pcs
2N7002K	SOT-23	72K	Reel, 3000 pcs

Document revision 1.0. Values are typical at TA = 25 degC. Verify against the latest datasheet from Diodes Incorporated before design release.